

PROMPTMINER-SAM2: PROMPT MINING FOR REMOTE SENSING INSTANCE SEGMENTATION

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ABSTRACT

In dense overhead scenes, visually similar adjacent instances are difficult to separate under shadows, low contrast, and narrow inter-instance gaps. Consequently, a proposal box alone provides insufficient shape cues for accurate instance-mask prediction. To address this issue, we propose PromptMiner-SAM2, a two-stage framework that mines proposal-level shape cues and injects them into prompt-conditioned SAM2 mask decoding. The mined shape cues are converted into complementary prompts that provide instance-specific guidance for SAM2 mask prediction. To address residual ambiguity in the decoded masks, an uncertainty-guided decoder-tail point refinement module further corrects selected uncertain locations. Experiments on the WHU and NWPU VHR-10 benchmarks show that PromptMiner-SAM2 obtains 73.4% and 68.3% mask AP, respectively, achieving the highest mask AP among the compared methods on both datasets.

Index Terms— Instance segmentation, remote sensing imagery, SAM2, prompt mining, mask refinement

1. INTRODUCTION

Overhead instance segmentation supports cadastral mapping, urban planning, and post-disaster assessment [1], [2]. In dense overhead scenes, visually similar adjacent instances are difficult to separate under shadows, low contrast, and narrow inter-instance gaps. Once candidate instances have been localized, a rectangular proposal box provides only a coarse spatial extent and insufficient shape cues for accurate instance-mask prediction. As illustrated in Fig. 1, box-only prompting can localize neighboring buildings but leaves their separating boundary under-specified, motivating richer instance-specific shape cues for mask prediction.

Existing remote-sensing instance segmentation methods couple instance discovery with mask prediction through two-stage, one-stage, cascade, or conditional designs [3]–[6]. Transformer-style frameworks provide another modeling route by using global sequence representations for set-based prediction [7]–[9]. Alternative instance representations have also been explored through diffusion-based filter denoising and architecture search for efficient prediction [18]–[19]. These approaches generally rely on rectan-

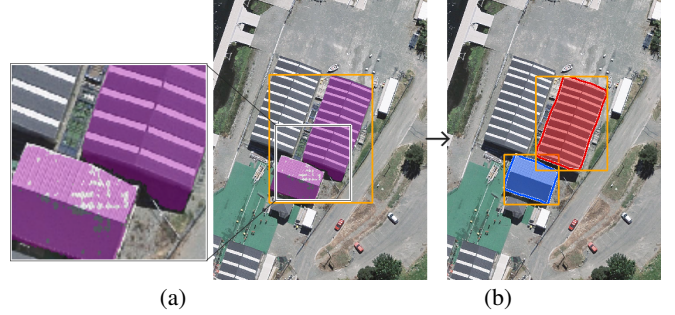


Fig. 1. An example of ambiguous instance separation caused by limited box cues. (a) A box-only prediction provides coarse localization but does not explicitly specify the separating boundary between neighboring buildings. (b) Ground-truth instance masks with distinct contours reveal the shape information required for accurate instance separation.

gular proposals or generic instance queries, which provide limited target-shape cues when neighboring objects have similar appearance. Beyond these architectures, prompt-conditioned foundation models offer a complementary way to inject spatial constraints into mask decoding. SAM and SAM2 accept sparse or dense prompts, motivating remote-sensing adaptation and automatic prompt learning such as RSPrompter [10]–[15]. SAMRefiner constructs multiple prompt types from an initial mask, while PointRend predicts at uncertain locations [16], [17].

These lines of work point to a common opportunity: prompt-conditioned decoders can benefit from instance-specific spatial cues beyond a rectangular extent. In a proposal-driven SAM2 pipeline, the detector already produces proposal-level features, but a box alone does not fully convey the shape of the localized instance for mask prediction. Preserving and transforming these cues into prompt-compatible guidance therefore becomes important for connecting instance discovery with accurate SAM2 mask decoding. To this end, we propose PromptMiner-SAM2, a two-stage framework that integrates proposal-based instance discovery with prompt-conditioned SAM2 segmentation.

Specifically, our contributions are threefold:

- We propose a proposal-driven two-stage framework for

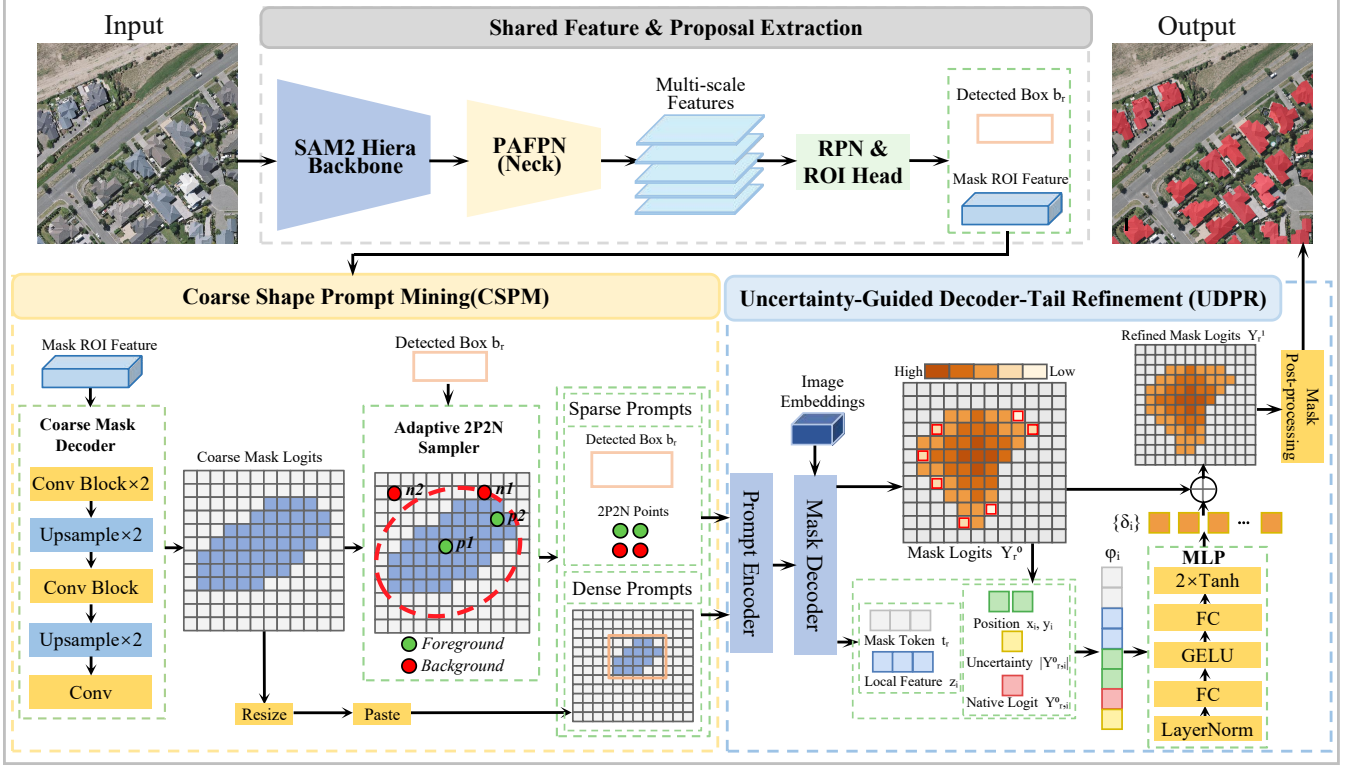


Fig. 2. Overview of the proposed PromptMiner-SAM2. SAM2 Hiera features support proposal extraction and provide image embeddings for mask decoding. Each detected box and its mask ROI feature drive coarse-shape prompt mining, which produces point, box, and dense-mask prompts. Uncertainty-guided decoder-tail refinement updates selected uncertain logits before mask post-processing.

remote-sensing instance segmentation, coupling SAM2 with detector-based proposal extraction to connect instance discovery with prompt-conditioned mask decoding.

- We propose the coarse-shape prompt mining module (CSPM), which converts proposal-level shape estimates into complementary prompts to preserve instance-specific cues for separating adjacent objects.
- We design the uncertainty-guided decoder-tail refinement module (UDPR), which targets ambiguous mask locations with localized residual corrections to improve instance-mask delineation.

2. METHOD

As depicted in Fig. 2, PromptMiner-SAM2 couples a two-stage detector with the SAM2 prompt-to-mask interface. Hiera features feed the PAFPN/RPN/ROI pathway for proposal extraction, while the SAM2 mask decoder combines the image embedding with proposal-conditioned prompts. The resulting decoder features and mask logits are refined at un-

certain locations, yielding the final instance masks for each detected proposal after post-processing.

2.1. Coarse-Shape Prompt Mining

CSPM converts each proposal’s ROI feature into a local shape estimate and uses it to construct complementary geometric cues for SAM2. It outputs 2P2N point prompts, the proposal box, and a dense shape prompt in the image-aligned coordinate frame.

For each proposal, the coarse branch maps its ROI feature X_r to a proposal-local shape logit map

$$C_r = f_{\text{coarse}}(X_r), \quad C_r \in \mathbb{R}^{1 \times 64 \times 64}. \quad (1)$$

Let $p_r = \sigma(C_r)$. PromptMiner derives a foreground support at probability 0.5 from p_r , while retaining the signed logits C_r for the dense prompt. It mines two positive and two negative points (2P2N) from this shared coarse shape. The first positive point is selected among confident foreground interior pixels ($p_r \geq 0.6$) by its foreground confidence weighted by its normalized distance to the support boundary; the second follows the same criterion while remaining spatially separated from the first. The first negative point comes from the

one-pixel outer ring with low foreground probability ($p_r \leq 0.3$). The second is selected from background outside a dilation of the foreground support and is separated from the first positive and ring-negative points.

These four points encode complementary interior, exterior, and background cues. The 2P2N set maps them to signed prompts in the image frame.

Coordinates map to the image frame; signed logits are resized and pasted into b_r 's mask-input canvas:

$$\mathcal{S}_r = \mathcal{M}(C_r, b_r), \quad Q_r = \text{Place}(C_r, b_r), \quad (2)$$

Here, $\mathcal{M}$ denotes 2P2N selection and coordinate mapping, and Place is the resize-and-paste operation. The two corners of b_r provide the box prompt. SAM2 jointly encodes the point, box, and dense-mask prompts, then decodes them with the image embedding in one pass:

$$Y_r^0 = \mathcal{D}(\mathcal{E}_I(I), \tilde{\mathcal{E}}_P(\mathcal{S}_r, b_r, Q_r)), \quad (3)$$

where $\mathcal{E}_I$, $\tilde{\mathcal{E}}_P$, and $\mathcal{D}$ denote the SAM2 image encoder (including its decoder-side high-resolution features), complete SAM2 prompt-embedding pathway, and mask decoder, respectively. The pathway $\mathcal{E}_P$ includes the SAM2 composition of the dense-mask embedding. Q_r retains the signed rather than binarized coarse logits. Thus, the dense prompt preserves the spatial variation of the proposal-local shape, while the point and box prompts provide discrete geometric constraints to the same decoder.

2.2. Uncertainty-Guided Refinement

UDPR focuses on ambiguous locations in the initial SAM2 mask. It combines local decoder features with instance-level context to predict localized residual corrections. The same decoding pass returns an upsampled decoder feature map U_r , the first output mask token t_r , and decoder logit map Y_r^0 . On the raster grid $\mathcal{G}_r$, UDPR visits the K_r locations with the smallest logit magnitudes:

$$\begin{aligned} K_r &= \min(K, |\mathcal{G}_r|), \\ \Omega_r &= \text{BottomK}(|Y_r^0|, K_r), \quad K = 64, \end{aligned} \quad (4)$$

where BottomK returns the K_r indices in $\mathcal{G}_r$ with the smallest logit magnitudes. For each $i \in \Omega_r$, the aligned feature z_i is gathered from U_r and combined with the token, normalized location, local logit, and logit magnitude:

$$\phi_i = [z_i, t_r, Y_{r,i}^0, x_i, y_i, |Y_{r,i}^0|]. \quad (5)$$

The local feature describes the decoder output near i , the mask token conditions the update on the decoded instance, and the coordinate and logit terms specify the position and current decision state. A lightweight MLP g then predicts a bounded residual for each selected location:

$$\delta_i = 2 \tanh g(\phi_i), \quad i \in \Omega_r, \quad (6)$$

The refined logit map is formed by scattering these residuals back to their original positions:

$$Y_{r,i}^1 = \begin{cases} Y_{r,i}^0 + \delta_i, & i \in \Omega_r, \\ Y_{r,i}^0, & i \notin \Omega_r. \end{cases} \quad (7)$$

This selective update leaves all nonselected decoder logits unchanged and applies bounded residual corrections on the existing SAM2 decoder grid.

3. EXPERIMENTS

3.1. Experimental Setup

Experiments use the WHU dataset [20], a single-class building benchmark with 2,943/627/2,220 training/validation/test images, and the NWPU VHR-10 dataset [21], a ten-class benchmark. Because the latter provides detection annotations only, we use the publicly released instance masks of Su *et al.* [22] and its 520/130 train/evaluation split.

Our PyTorch implementation uses distributed data parallelism on four NVIDIA RTX 3090 GPUs, with one image per GPU and two-step gradient accumulation (effective batch size eight). Both benchmark runs initialize from SAM2 Hiera-Base+ weights: base Hiera parameters remain frozen, while its LoRA adapters and the mask decoder are updated. We use AdamW with an initial learning rate of 5×10^{-4} and cosine annealing; training uses cross-entropy, Smooth L1, binary cross-entropy, and Dice losses. A value of $K = 64$ is selected by screening $K \in \{32, 64, 128\}$ under a common protocol; it achieves the highest overall bbox AP and mask AP in Table 1 and is used in the main experiments.

Table 1. Sensitivity of K on the NWPU VHR-10 dataset under a common development protocol.

K	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
32	69.38	91.69	79.03	64.94	92.91	70.01
64	70.35	92.23	81.65	66.65	94.15	71.97
128	70.13	92.42	81.32	65.97	93.71	72.22

3.2. Comparative Analysis

PromptMiner-SAM2 is compared with 10 representative instance-segmentation methods. They are grouped into CNN-based methods (CNN) and global sequence modeling methods (GSM). CNN includes two-stage, one-stage, and cascade designs, whereas GSM includes Transformer-style mask prediction and foundation-model prompting. The *Type* column in Table 2 uses these two abbreviations.

It can be concluded from Table 2 that our method obtains the highest mask AP on the WHU dataset and the NWPU VHR-10 dataset, leading by 0.1 and 2.2 points, respectively,

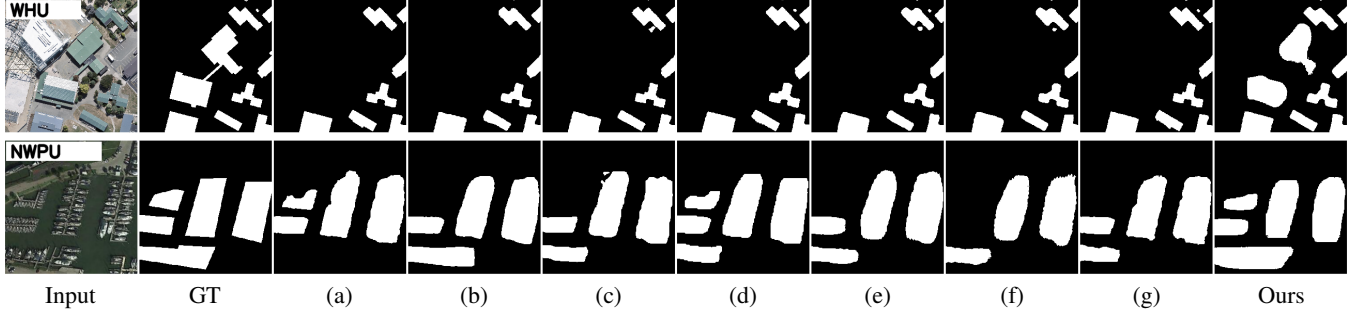


Fig. 3. Qualitative comparison on the WHU and NWPU VHR-10 datasets. From left to right, the columns show the input image, ground truth, predictions of (a) MaskDINO, (b) RSPrompter, (c) CondInst, (d) YOLO11, (e) HTC, (f) Mask R-CNN, (g) M2F, and PromptMiner-SAM2 (ours).

Table 2. Comparative analysis on the WHU dataset and the NWPU VHR-10 dataset. Each entry reports AP / AP₅₀ / AP₇₅ (%) for the Bbox and Mask columns.

Type	Method	WHU						NWPU VHR-10					
		Bbox (%)			Mask (%)			Bbox (%)			Mask (%)		
		AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
CNN	Mask R-CNN [3]	70.9	89.9	80.9	68.2	90.0	80.0	65.4	90.2	78.9	63.0	90.1	71.2
	MS R-CNN [4]	68.8	89.4	80.0	67.7	89.6	79.5	67.8	94.3	83.8	64.7	93.6	80.8
	HTC [5]	72.7	90.6	82.5	69.2	90.7	81.6	58.9	86.9	67.7	51.3	84.6	53.3
	CondInst [6]	74.2	90.0	82.4	69.7	90.2	80.9	65.0	90.1	79.6	62.8	90.7	68.1
	YOLO11s-seg [23]	76.7	91.2	85.4	66.8	91.3	79.1	71.0	94.4	79.5	63.9	93.9	67.3
GSM	Mask2Former [7]	61.4	82.7	70.7	64.5	84.1	75.6	60.3	77.9	66.8	61.5	84.9	67.8
	MaskDINO [8]	73.3	90.8	82.7	73.3	90.3	82.5	65.7	88.9	74.2	65.8	89.6	71.5
	RS4D-box [9]	64.5	87.6	74.7	64.1	87.8	75.2	57.0	82.5	65.4	54.3	80.1	57.6
	RSIISN [1]	74.1	90.0	84.4	72.1	90.1	83.9	66.7	91.5	78.4	62.9	89.7	67.1
	RSPrompter [13]	72.5	91.0	81.7	72.5	92.0	82.9	70.3	93.6	81.0	66.1	92.7	70.6
Ours	PromptMiner-SAM2	76.4	91.5	85.4	73.4	91.4	85.5	73.4	94.0	84.2	68.3	94.5	74.6

over the strongest alternatives. Its WHU mask AP₇₅ is also the best, while the bbox results remain close to the leading YOLO11s-seg baseline. On the NWPU VHR-10 dataset, it leads the bbox AP, bbox AP₇₅, mask AP, and mask AP₅₀ comparisons; although mask AP₇₅ is below the best value, it remains competitive with other GSM methods.

Table 3. Prompt-path ablation on the NWPU VHR-10 dataset; *P*, *B*, and *M* denote point, box, and dense-mask prompts, and UDPR denotes uncertainty-guided refinement.

Variant	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
<i>P</i>	71.81	93.60	81.89	63.15	92.56	67.73
<i>B</i>	72.14	94.34	82.24	65.25	94.12	69.02
<i>P+B</i>	71.74	93.52	81.68	66.94	93.93	72.73
<i>P+B+M</i>	73.76	93.80	84.69	66.35	93.66	71.61
<i>P+B+M</i> + UDPR	73.43	93.96	84.16	68.28	94.53	74.61

3.3. Ablation Study

The effects of adding prompt types and decoder-tail refinement are summarized in Table 3. With only the box prompt *B*, mask AP is increased from 63.15% to 65.25%, and mask AP₇₅ from 67.73% to 69.02%. With both discrete prompts (*P+B*), 66.94% mask AP and 72.73% mask AP₇₅ are ob-

Figure 3 compares predictions on the WHU dataset and the NWPU VHR-10 dataset. In the WHU row, touching roofs remain separated; in the NWPU row, object silhouettes are more complete with fewer merges.

tained, showing complementary cues.

With the dense-mask prompt, 66.35% mask AP and 71.61% mask AP_{75} are obtained, while bbox AP and bbox AP_{75} are 73.76% and 84.69%, respectively. For the complete $P+B+M+UDPR$ configuration, 68.28% mask AP and 74.61% mask AP_{75} are obtained, the best mask results in this ablation.

4. CONCLUSION

This paper presents PromptMiner-SAM2, a framework that addresses ambiguous instance separation in overhead imagery by converting proposal-local shape cues into complementary prompts for SAM2 and refining uncertain mask locations. The resulting design improves remote-sensing instance segmentation, reaching 73.4% mask AP on the WHU dataset and 68.3% on the NWPU VHR-10 dataset. Future work can explore richer prompt-mining schemes, including cues that adapt to object scale and local scene context.

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